Low-dimensional coexistence in complex microbial ecosystems

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Niche theory predicts niche dimensionality to drive diversity

- Current physical models treat microbiomes as disordered systems with random, high-dimensional interactions (akin to spin glasses).
- This core assumption of high-dimensional coexistence remains untested empirically.
- Niche theory predicts higher niche dimensionality, η_D, drives diversity by reducing niche overlap, but estimating η_D is difficult.
- We take an effective approach to estimate η_D directly from abundance profile data.

figures/Niche overlap blobs.jpg

Pianka, Eric R. Evolutionary ecology. Eric R. Pianka, 2011.

Latent variable JSDMs can be fit to data and is interpretable

- JSDMs relate species distribution to environmental factors and estimate η_D.
- Latent variable formulization of JSDMs decompose microbiomes into interpretable sample specific and species specific embeddings.
- We define η_D as the characteristic environmental dimensionality that explains composition.

figures/model as matrix factorization.p

figures/model interpretability.png

Consumer resource models are mechanistic and widely used

- The consumer resource framework uses environmental mediation of bacterial interaction to model microbiomes.
- We incorporate two types of dimensional constraints that can theoretically allow species to exist in low-dimensional niches.

$$\frac{1}{n_o}\frac{dn_o}{dt} = \sum_{k=1}^{K} r_k(t)\theta_{ko} - \delta_o$$

$$\frac{dr_k}{dt} = J_k(t) - \sum_{o=1}^{O} r_k(t)\frac{\theta_{ko}}{w_k}n_o(t)$$

figures/Model.png

- **Intrinsic tradeoffs:** Modeled as simplex constraints on species preference vectors, placing them on equal footing.
- **Extrinsic tradeoffs:** Modeled as low-rank correlations in nutrient inflow vectors that reduce the effective degrees of freedom in the environment.

Tradeoffs explains how η_D relates with community structure

- Tradeoffs successfully reproduce the empirical relationships between community structure, niche overlap, and niche dimensionality.

Tradeoffs explain low η_D of microbiomes

- Across ~ 200 real microbiome datasets obtained from the Microbiomap project we see that microbiomes are highly low dimensional.
- **Disordered consumer-resource models tuned to match empirical diversity predict median $\eta_D \approx 13$, directly contradicting the empirically observed median $\eta_D \approx 4$.**
- In simulations, extrinsic tradeoffs (correlated resource supplies) impose stronger dimensional constraints on the effective niche space than intrinsic metabolic tradeoffs alone.

figures/BoxPlot Niche dimensionality.png

Implications for Statistical Physics Models

- Our empirical estimates directly challenge theoretical models that draw interactions or consumer preferences i.i.d. from fixed distributions.
- Coexistence in complex microbial ecosystems is not governed by effectively infinite, independent ecological axes.
- To accurately capture macroscopic ecological dynamics, future statistical physics models must explicitly incorporate dimensional constraints.
- **Key Takeaway:** Incorporating low-rank interaction matrices (in Generalized Lotka-Volterra systems) and correlated resource structures (in CRMs) is essential to model realistic microbiome phase spaces.

References

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